

UNIT - I
2 MARKS QUESTIONS & ANSWER
CRYSTAL STRUCTURE & DEFECTS

1) What is a space lattice ?

Space lattice is an infinite arrangement of points in three dimensions in which every point has an identical surrounding.

2) What is a Unit cell ? (or) Define Unit cell ?

Unit cell is defined as the basic unit or the smallest volume from which the entire crystal is built up by translational repetition in three dimensions (OR) A unit cell is the smallest unit which, when repeated in space indefinitely, will generate the space lattice.

3) Name seven crystal systems ?

i) Cubic ii) Tetragonal iii) Orthorhombic iv) Monoclinic v) Triclinic vi) Rhombohedral
vii) Hexagonal.

4) Explain Bravais lattice? (OR) What are Bravais lattice?

Bravais lattices are the 14 types of space lattices in the 7 systems of crystals.

i) 3-Cubic ii) 2-Tetragonal iii) 4-Orthorhombic iv) 2-Monoclinic v) 1-Triclinic
vi) 1-Rhombohedral vii) 1-Hexagonal.

5) Define primitive cell ?

The unit cell formed by the primitives a , b , and c having only lattice point is called the primitive cell.

Example: Simple cubic unit cell, existing in polonium.

6) Define coordination number?

Coordination number is the number of nearest neighbor atoms that an atom has in the given crystal structure at a distance $2r$ where r is the atomic radius.

7) What are the coordination numbers in FCC, and HCP?

Type of Structure	Coordination numbers
Face Centred Cubic	12
Hexagonal Closed Packed	12

8) Give the values of number of atoms in unit cell of BCC and HCP?

Type of Structure	Number of atoms in unit cell
Face Centred Cubic	4
Hexagonal Closed Packed	6

9) Define atomic radius?

The half the distance between nearest neighbouring atoms in a crystal is known as atomic radius. The atomic radius is denoted by r and is usually expressed in terms of the cubic edge a (lattice parameter).

12) Derive an expression for atomic radius r in terms of lattice constant for FCC?

The length of the face diagonal = $4r$ where r is the atomic radius.

From the lattice parameter, the length of the face diagonal = $\sqrt{2a^2} = \sqrt{2} a$

$$\text{(or) } 4r = \sqrt{2} a \quad r = \frac{a}{2\sqrt{2}}$$

13) Define packing factor .What is its unit?

Packing factor is the ratio between the volume occupied by the atoms in the unit cell v and total volume of the unit cell V .

$P.F = v/V$ There is no unit for it since it is a ratio.

14) Write down the packing factor of FCC crystal ?

$$\text{Packing factor} = \frac{4 \times \frac{4}{3} \pi r^3}{a^3}$$

$$= \frac{16\pi}{3a^3} \left[\frac{a}{3\sqrt{2}} \right]^3$$

$$= \frac{\pi}{3\sqrt{2}} = 0.74$$

15) State c/a ratio and packing factor of HCP?

$$c/a = \sqrt{\frac{8}{3}} = 1.633$$

$$c/a = 1.6333$$

$$\text{P.F} = \frac{\pi}{3\sqrt{2}} = 74\%$$

16) What are Miller indices ?

Miller indices are 3 possible integers which have same ratios as the reciprocals of the intercepts of the plane concerned on the 3 axes.

17) Write down the expression for inter planar spacing for a cubic system in terms of lattice constant?

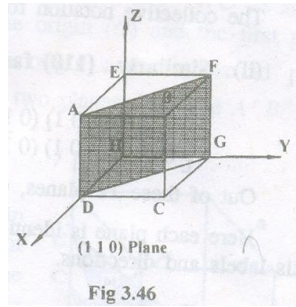
$$\text{Inter planar spacing } d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Where a = lattice constant and h, k, l = Miller indices.

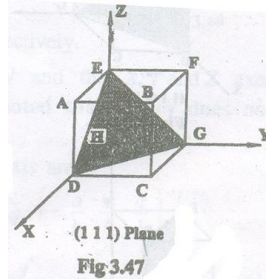
18) How do we indicate the normal to the plane of (h k l) in symbols?

The direction [h k l] is the normal to the plane (h k l).

19) Sketch the (110) plane in a cubic system?



20) Sketch the (111) plane in a cubic crystal?



21) Lattice constant of copper is 0.38 nm. Calculate the distance between (110) plane?

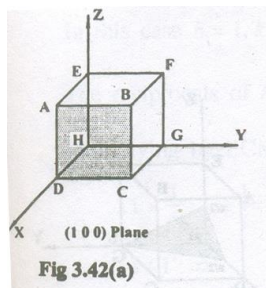
$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$= \frac{0.38 \times 10^{-9}}{(1^2 + 1^2 + 0^2)^{\frac{1}{2}}}$$

$$= \frac{0.38 \times 10^{-9}}{\sqrt{2}}$$

$$= 0.27 \text{ nm}$$

22) Sketch the (100) plane in a cubic crystal?



23) Explain Procedure for finding Miller indices? (OR) What are the steps for finding Miller indices?

- i) Find the intercepts of the plane along the coordinate axes X,Y,Z.
- ii) Take the reciprocal of these intercepts.
- iii) Reduce the reciprocals into whole numbers.
- iv) Write these integers within parentheses to get Miller indices

24) What is meant by loosely packed crystal structure ? Give an example of this types of material?

The loosely packed crystal structure has the packing factor less than 0.74 .That is ,in which more vacant site is available .Simple cubic polonium .body centred sodium and diamond cubic silicon have these loosely packed crystal structure.

25) What is meant by closely packed crystal structure ? Give an example of this types of material?

Closely packed crystal structure has the highest packing factor of 0.74 .Here the atoms are atoms are closely packed leaving a small space as vacant site in the crystal .FCC copper and hexagonal closed packed magnesium belong to this closely packed crystal structure.

26) What is a crystal?

A crystal is a solid composed of a periodic arrangement of atoms.

27) What are the lattice parameters of an unit cell?

The characteristic intercepts on the axes a,b,c and interfacial angles are called lattice points.

28) What are lattice points?

The points in the space lattice are called lattice points.

29) What is basis?

The crystal structure is formed by associating with every lattice point a unit assembly of atoms or molecules .This unit assembly is called the basis or pattern.

30) What are the different between Crystals and Amorphous solids?

Crystals	Amorphous solids
In each crystal, the atoms or molecules are arranged in a regular and orderly in a 3D pattern, called lattice.	In an amorphous solid, molecules which are the basic units are arranged in an irregular manner, otherwise there is no lattice structure.
They possess internal spatial symmetry of atomic or molecular orientation.	These do not possess any internal spatial symmetry.
When a crystal breaks, the broken pieces are all having regular shape.	When an amorphous solid breaks, the broken pieces are irregular in shape.
The crystal have directional properties and therefore called anisotropic substances.	These solids have no directional properties and therefore called isotropic substances.
Examples : i)Metallic crystals :Copper, Silver, Aluminium , etc., ii)Non-metallic crystals :Diamond, Silicon, Germanium, Sodium Chloride, etc.,	Examples: Glass , Plastics ,Rubber , etc .,

31) Mention few applications of crystal imperfections or point defects ?

i) Doped atoms or impurities in silicon or germanium crystal increase its electrical conductivity.

ii) The interstitial impurity carbon atoms in iron lattice increase the mechanical strength of iron. Brittle, soft iron is converted into high strength hard steel.

iii) Addition of copper atoms in gold increases the ductility of gold.

Iv) In copper lattice, the substitutional impurity atoms of tin increases the bearing properties of copper.

32) What are dislocations? What are their uses?

A dislocation is defined as a disturbed region between 2 substantially perfect parts of a crystal and is an 1D crystal imperfection . They are used to produce slip in plastic flow during mechanical working of a material .Further these are used to get superconductivity at higher transition.

33) What is meant by imperfection in a crystal?

Any deviation in a natural crystal from the regular and the periodic arrangement of atoms is known as crystal imperfection.

34) Explain edge dislocation?

Edge dislocation line can be considered as the defect due to the insertion of an extra half plane of atoms. If this extra plane of inserted atoms is above the slip plane the line defect is called positive dislocation and it is denoted by the symbol $\rightarrow$. Instead, if this extra plane is below, the defect is called negative dislocation and is denoted by the symbol $\leftarrow$.

35) What is a point defect?

The defects which take place due to imperfect packing of atoms during crystallization are known as point defects.

36) What is a line defect?

The defects which take place due to dislocation or distortion of atoms along a line in some direction is called as line defect.

37) What is meant by Surface imperfections or surface defects or plane defect?

The defects, which take place on the surface of a material are known as surface defects or plane defects. The surface defects take place either due to imperfect placing of atoms during crystallization or defective orientation of the surface.

38) Explain the Screw dislocation ?

Screw dislocation results from a displacement of the atoms in one part of a crystal relative to the rest of crystal forming a spiral ramp around the dislocation line.

39) What is meant by Burger vector?

The vector which indicates the direction and magnitude of the shift of the lattice on the slip plane is called a Burger vector.

40) What is a Schottky defect?

Whenever a pair of positive and negative ions are missing from a crystal the defect is known as schottky defect . i.e., the pair of ion vacancies is termed as schottky defect. In this type of defect the crystal is electricity neutral.

41) What is a Frenkel defect?

An atom may leave its regular site and may occupy nearby interstitial site of the matrix giving rise to two defects simultaneously i.e., one vacancy and the other self interstitial. These two defects together is called Frenkel defect.

42) What is meant by grain boundary?

Grain boundaries are the defects which separate grains of different orientation from each other in a polycrystalline material.

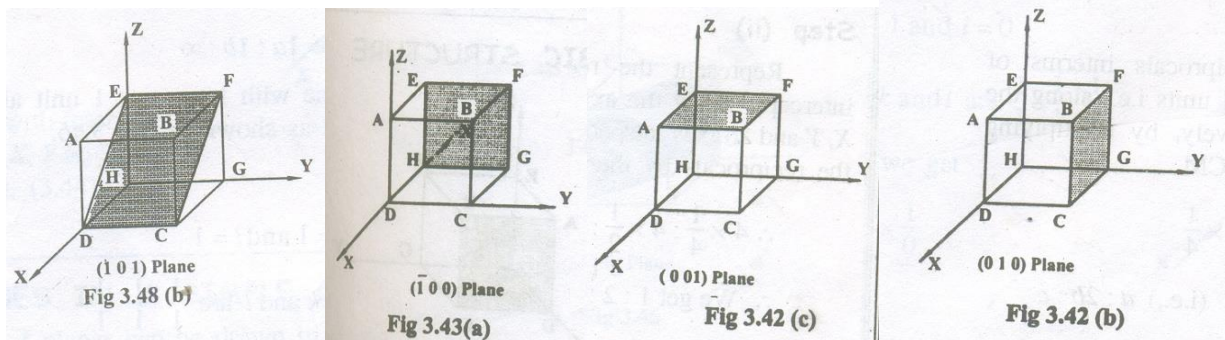
43) Explain tilt boundary?

Tilt boundary is another type of surface defect. It is called a small angle boundary as the orientation of grains in two neighbouring crystals differ by only a few degrees i.e., less than 10 degree.

44) Explain twin boundary?

Twin boundary are the boundaries in the grains at which the atomic arrangement on one side of the boundary is the mirror image of the atom on the other side.

45. Stretch the planes (101), (100), (001) and (010) planes in a simple cubic cell.



Prepared By: Dr. K.B. Renukadevi, HOD, Dept. of Physics, RGCET